

3. LITERATURE REVIEW

3.1 OBSOLESCENCE OF SIGNATURE-BASED IDS

Signature-based IDS are capable of detecting only known attacks and cannot detect most new OT threats (Stallings, 2023). The NCSC (2023) confirms that state-sponsored APT actors are routinely creating custom tooling to bypass known rule sets. In a recent report, Ahmed (2023) highlights OT-targeted ransomware groups scheduling payload delivery based on OT handover operations tactic that takes advantage of the signature-based detection gap. The main survey supports this empirically: Q4 mean=2.32/5, 68.2% scored 1 or 2.

3.2 AI AS TRANSFORMATIVE DEFENCE

Gupta (2023) provides the most comprehensive academic treatment of deep learning applied to intrusion detection, with LSTM autoencoders reducing MTTD by up to 63% in enterprise IT environments. According to Darktrace (2024), the mean-time-to-response is 87% faster. Both lack validation of their performance against real OT traffic, which is addressed by this study's SWaT and CIC-IDS benchmarking. XGBoost was proposed by Chen and Guestrin (2016); benchmarking shows that it is out-performed by autoencoders for novel OT anomaly detection, but competitive for tabular data. Class imbalance is one of the most important problems in intrusion detection datasets, and SMOTE (Chawla et al., 2002) was used to solve this problem.

3.3 EXPLAINABLE AI (XAI)

Schneiderman (2020) holds that in safety-critical settings, human-centered AI design is a moral duty. Unexplainable AI alerts result in alert fatigue, override, and loss of trust. SHAP and LIME give post hoc explanations for individual predictions. The survey confirms: Q9 mean=3.50/5; Mode=5; 54.5% score 4–5; 59.1% expect XAI to increase operator trust (Q10).

3.4 FEDERATED LEARNING

FL (McMahan et al., 2017) enables organizations to train common models without sharing raw data. Kizza (2023) identifies governance and liability fragmentation as principal FL barriers directly corroborated: 81.8% are FL-aware (Q11) but only 27.3% are willing to participate (Q12). To protect FL from model poisoning attacks, Byzantine fault-tolerant aggregation is required, as pointed out by Mehrabi et al. (2022).

3.5 LITERATURE GAP ANALYSIS

Source	Key Finding	Gap Identified	This Study Addresses?
Stallings (2023)	Signature IDS miss >50% novel OT threats	No OT latency benchmarks	Yes algorithm benchmarking
Gupta (2023)	DL reduces MTTD 63% in enterprise	No OT validation	Yes Swat/CIC-IDS testing
Darktrace (2024)	87% faster MTTR enterprise	CI/OT evidence absent	Yes OT simulation
Shneiderman (2020)	Human oversight essential in AI	No XAI framework for OT	Yes XAI survey analysis
Kizza (2023)	Governance gap primary CI barrier	No FL model for CI	Yes FL survey + analysis
Ahmed (2023)	Ransomware now OT-targeted	No AI countermeasure evaluation	Yes adversarial testing
NCSC (2023)	APT evasion of known rules	No AI performance benchmark	Yes comparative analysis
Chen & Guestrin (2016)	XGBoost: scalable tree boosting	No OT anomaly detection context	Yes baseline comparison

Source	Key Finding	Gap Identified	This Study Addresses?
Chawla et al. (2002)	SMOTE: class imbalance correction	Not applied in OT IDS context	Yes applied to all supervised models
McMahan et al. (2017)	FL enables privacy-preserving learning	No CI governance framework	Yes FL governance analysis